

Self-Powered Sensing via Piezoelectric Energy Harvesting

Executive Technology Brief

Domain: Piezoelectric Energy Harvesting + Implantable Sensing

Technology Maturity: ENGINEERING_DEFINITION

Transfer Posture: Sponsored Validation

Problem: Chronic ICP monitoring requires battery or percutaneous leads

Mechanism: Piezoelectric transducer harvests energy from CSF pulsation; duty-cycled sensing matches harvested power budget (R1 fix)

Architecture: Self-powered CSF pressure/flow sensor powered by harvested mechanical energy from CSF pulsations or catheter motion. R1 fix: original P-15 assumed unrealistic power density; R1 redesign uses piezoelectric transducer with rectifier + storage + duty-cycled sensing.

Governing Model: Piezoelectric energy harvesting from low-frequency mechanical strain. Power output scales with stress squared, active volume, and material coupling coefficient. Duty-cycled sensing matches available power.

Key Equations:

$V = g_{33} * d * \sigma$ [piezoelectric voltage]

$P \sim (d_{33} * g_{33} * \sigma^2 * Volume) / (2 * duty_cycle)$ [piezo harvested power]

$P_{avg} = (d_{33} * g_{33} * \sigma_{rms}^2 * Volume * \eta) / (2 * T_{cycle})$

$E_{stored} = 0.5 * C * V^2$ [capacitor storage]

$f_{duty} = P_{harvested} / P_{sensor_active}$ [achievable duty cycle]

Kill Condition: Harvested power at in-vivo strain levels is insufficient for any duty cycle (model shows <0.1 uW average)

Current State: ENGINEERING_DEFINITION. No physical prototype. No clinical validation.

Disclosure: This dossier distinguishes established evidence from proposals and unresolved questions.